

FURTHER CONDENSATION IDEAS DRAFT

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1. INTRODUCTION

1.1. Acknowledgments. Thanks to all that was and will be for providing the criteria of intelligibility and value of this paper.

2. BACKGROUND AND NOTATION

We will at times need to use measure theoretic hypotheses. We choose to do this in a simple way—in most cases, we assume that the measurable spaces that we work with are standard Borel. This class of spaces was introduced in Mackey (1957), and is discussed from a synthetic perspective in Fritz (2020). *Should I get better references, since Fritz just isn't clear enough about what's stated?* We briefly state some facts about such spaces.

Definition 2.1. A Polish space is a topological space that has the topology of some separable complete metric space.

Many familiar spaces are Polish, such as topological manifolds and separable Banach spaces. We can regard a Polish space Ω as a measurable space by taking its underlying set together with the collection $\mathcal{B}$ of all its Borel subsets. In other words, $\mathcal{B}$ is the smallest class of subsets of Ω containing all its open subsets and closed under the operations of complementation and countable union.

Definition 2.2. A standard Borel space is a measurable space $(\Omega, \mathcal{A})$ that is isomorphic as a measurable space to a Borel subspace of a Polish space.

Standard Borel spaces are especially convenient because of the existence of conditionals. For example, we have the following, adapted from Sections 3, 4, and 11 of Fritz (2020) and from Giry (1982). *This should all be in an appendix if its used at all, except maybe the theorem on disintegration.*

Given a measurable space Ω , the *measurable space of probability measures on Ω* , denoted $G(\Omega)$, is the measurable space whose underlying set is the set of all probability measures on Ω , and which is equipped with the smallest σ -algebra such that for all measurable $E \subseteq \Omega$, the function $G(\Omega) \rightarrow \mathbb{R}$ given by $\mathbf{P} \mapsto \mathbf{P}(E)$ is measurable (with respect to the Borel σ -algebra on $\mathbb{R}$).

We can equivalently express this as follows. Let $\mathcal{A}$ be the σ -algebra of measurable subsets of Ω , and let Λ be another measurable space. A function $f: \Lambda \rightarrow G(\Omega)$ naturally corresponds to a function $\tilde{f}: \Lambda \times \mathcal{A} \rightarrow \mathbb{R}$;

$$(2.1) \quad \tilde{f}(\lambda, E) = f(\lambda)(E).$$

Then, a function $\tilde{f}: \Lambda \times \mathcal{A} \rightarrow \mathbb{R}$ corresponds to a measurable function $\Lambda \rightarrow G(\Omega)$ if and only if (1) $f(\lambda, \cdot)$ is a probability measure for each choice of $\lambda \in \Lambda$ and (2) $f(\cdot, E)$ is a measurable function $\Lambda \rightarrow \mathbb{R}$ for each choice of $E \in \mathcal{A}$.

Proposition 2.3. *There is a monad G , the Giry monad, on the category of standard Borel spaces and measurable maps, which sends each standard Borel space Ω to the space $G(\Omega)$ (which is particular is indeed a standard Borel space), and sends each measurable map $f: \Omega \rightarrow \Lambda$ to the usual pushforward*

$$(2.2) \quad G(f)(\mathbf{P}) = f_*\mathbf{P}.$$

We can use this concept to give a measure-theoretic treatment of conditional probability, using the following disintegration theorem.

Theorem 2.4 (disintegration). *Let $(\Omega, \mathbf{P})$ be a standard Borel probability space, let Λ be a measurable space, and let $X: \Omega \rightarrow \Lambda$ be a random variable. Then, there is a measurable function $\mathbf{Q}: \Lambda \rightarrow G(\Omega)$ that behaves like a conditional probability in that*

$$(2.3) \quad \mathbf{P}(A) = \int_{\Lambda} \mathbf{Q}(\lambda, A) \, dX_*\mathbf{P}(\lambda) = \int_{\Omega} \mathbf{Q}(X(\omega), A) \, d\mathbf{P}(\omega)$$

(The measure $X_*\mathbf{P}$ is known as the distribution of X .) Further, this function is unique up to a null set, in that any two functions $\mathbf{Q}_1, \mathbf{Q}_2: \Lambda \rightarrow G(\Omega)$ with this property must agree on a set of full measure under $X_*\mathbf{P}$.

*****Should I cite Kallenberg? Should I spell out the equivalence between this and Kallenberg's Thm 5.3? Should I switch the order of this theorem and the following definition?

Definition 2.5. In the context described in this theorem, any function $\mathbf{Q}: \Lambda \rightarrow G(\Omega)$ satisfying the stated property is called a *regular conditional probability distribution* on Ω given X . We also use the notation

$$(2.4) \quad \mathbf{P}(A \mid X = \lambda) = \mathbf{Q}(\lambda, A)$$

for such a function. Here, the symbol $\mathbf{P}$ identifies which probability measure we are taking the conditional distribution of, and the symbol X identifies which random variable we are taking the conditional distribution with respect to. There are two things to be careful about here. First, $\mathbf{P}(\cdot \mid X = \cdot)$ is not determined by the data, but is only determined almost everywhere; when we use this notation, we are

picking one such function. So, this notation is most suited for situations where the choice does not matter, since we only use this function in ways invariant to this choice. Second, we must not interpret this notation as asserting that $\mathbf{P}(A \mid X = \lambda)$ is function only of the measurable sets A and $X = \lambda$. We have used the random variable X in the construction, not merely the set $\{\omega \mid X(\omega) = \lambda\}$.

We also have the following convenient proposition.

Proposition 2.6. *Let $(\Omega, \mathbf{P})$ be a standard Borel probability space, let X be a random variable on Ω , and let $\mathbf{P}(\cdot \mid X = x)$ be a regular conditional probability distribution on Ω given X . Then, for almost all $x \in \mathcal{R}X$ in the sense of the distribution $X_*\mathbf{P}$, we have $\mathbf{P}(X = x \mid X = x) = 1$.*

Proof. *****This is probably in Blackwell & Dubins... $\square$

In addition, standard Borel spaces have nice measurability properties—working with them will let us ensure that the sets and functions we want to consider are measurable. Notably, by Theorem 3.2 of Mackey (1957), an injective measurable function between standard Borel spaces has Borel image, and is a measurable isomorphism of its domain with its range. Further, the class of standard Borel spaces have rich closure properties, so that many spaces we may want to construct are still standard Borel. For example, countable sums and products of standard Borel spaces are standard Borel by Theorem 3.1 of Mackey (1957), and the space of probability measures on a standard Borel space is given the structure of a standard Borel space in the construction of the Giry monad (Giry, 1982). We can say a bit more about sums and products. We will use an additional fact. Every standard Borel space is isomorphic as a measurable space not just to a Borel subspace of a Polish space, but in fact to a Polish space by Corollary 13.4 of Kechris (1995).

Proposition 2.7. *Let $(\Omega_i)_{i \in I}$ be a finite or countably infinite family of Polish spaces. Then, the standard Borel space induced by the product space $\prod_{i \in I} \Omega_i$ is isomorphic to the product in the category of measurable spaces of the standard Borel spaces induced by each space Ω_i , via the natural map on their underlying sets. The same holds for the topological and measurable sums.*

Proof. Let M be the functor from the category of topological spaces to that of measurable spaces that sends each topological space to the measurable space with the same underlying set in which the measurable sets are exactly the Borel sets, and sends each continuous map to the same function, which is also a measurable map. In the topological category, we have continuous maps $\prod_{i \in I} \Omega_i \rightarrow \Omega_j$ for each $j \in I$, so we have corresponding maps in the measurable category

$$(2.5) \quad M \prod_{i \in I} \Omega_i \rightarrow M \Omega_j.$$

Using the universal property of the product, we have a measurable map

$$(2.6) \quad M \prod_{i \in I} \Omega_i \rightarrow \prod_{j \in J} (M \Omega_j),$$

which acts in the obvious way on their underlying sets. This map is injective so, as stated above, it is a measurable isomorphism. The dual argument using the universal property of sums gives the corresponding conclusion. $\square$

Proposition 2.8. *Let Ω be a standard Borel space. Then, the diagonal is measurable in the product measurable space $\Omega \times \Omega$.*

Proof. Pick some Polish space X whose Borel measurable space is Ω . We know that the diagonal is closed in $X \times X$, so it is measurable in $\Omega \times \Omega$. $\square$

Proposition 2.9. *Let Ω and Λ be standard Borel spaces. Then, the map $G(\Omega) \times G(\Lambda) \rightarrow G(\Omega \times \Lambda)$ that takes two probability measures to their product measure is a measurable map.*

Proof. First, let $\pi_\Omega: \Omega \times \Lambda \rightarrow \Omega$ be given by $\pi_\Omega(\omega, \lambda) = \omega$. By functoriality, the pushforward $G(\Omega \times \Lambda) \rightarrow G(\Omega)$ is measurable. The same holds for the other projection $\pi_\Lambda: \Omega \times \Lambda \rightarrow \Lambda$, so using the universal property of the product, the map $f: G(\Omega \times \Lambda) \rightarrow G(\Omega) \times G(\Lambda)$ given by

$$(2.7) \quad f(\mathbf{P}) = ((\pi_\Omega)_* \mathbf{P}, (\pi_\Lambda)_* \mathbf{P})$$

is measurable.

Next, we show that the set $S \subseteq G(\Omega \times \Lambda)$ consisting of all those probability measure that are products of a probability measure on Ω and one on Λ is a measurable set. The first step is to give an alternative characterization of which such measures are products. Since Ω and Λ are standard Borel, we can pick Polish topologies that give rise to their measurable structures, and then there are families of sets $\{E_n\}_{n=1}^\infty$ and $\{F_n\}_{n=1}^\infty$ that are countable bases for the respective topologies on Ω and Λ . Now, if $\mathbf{P} \in G(\Omega \times \Lambda)$ is a product, then

$$(2.8) \quad \mathbf{P}(E_m \times F_n) = \mathbf{P}(E_m \times \Lambda) \mathbf{P}(\Omega \times F_n)$$

for all $m, n \in \mathbb{N}$. Conversely, suppose that the equation

$$(2.9) \quad \mathbf{P}(E_m \times F_n) = \mathbf{P}(E_m \times \Lambda) \mathbf{P}(\Omega \times F_n)$$

holds for all $m, n \in \mathbb{N}$. Let $\mathcal{A}$ be the largest family of measurable subsets of Ω such that

$$(2.10) \quad \mathbf{P}(A \times F_n) = \mathbf{P}(A \times \Lambda) \mathbf{P}(\Omega \times F_n)$$

for all n , and let $\mathcal{B}$ be the largest family of measurable subsets of Λ such that

$$(2.11) \quad \mathbf{P}(A \times B) = \mathbf{P}(A \times \Lambda) \mathbf{P}(\Omega \times B)$$

holds for all $A \in \mathcal{A}$. Since $\mathbf{P}$ is a measure, it follows that $\mathcal{A}$ and $\mathcal{B}$ are closed under countable union and complementation, so each is the σ -algebra generated by the corresponding family, $\{E_m\}_{m=1}^\infty$ or $\{F_n\}_{n=1}^\infty$. But these families were chosen so as to generate the entire σ -algebra on the respective measurable space, so in fact equation (2.11) holds for all measurable $A \subseteq \Omega$ and $B \subseteq \Lambda$. By definition, sets of the form $A \times B$ for measurable A and B generate the σ -algebra on $\Omega \times \Lambda$, so $\mathbf{P}$ is equal to the product of $(\pi_\Omega)_* \mathbf{P}$ and $(\pi_\Lambda)_* \mathbf{P}$, since it is equal on a generating set.

Now we know that S is the set of all measures in $G(\Omega \times \Lambda)$ satisfying the countable set of conditions (2.8). To confirm that S is measurable, we need only show that each of these sets is measurable. But the functions

$$(2.12) \quad \mathbf{P} \mapsto \mathbf{P}(E_m \times F_n)$$

$$(2.13) \quad \mathbf{P} \mapsto \mathbf{P}(E_m \times \Lambda)$$

$$(2.14) \quad \mathbf{P} \mapsto \mathbf{P}(\Omega \times F_n)$$

are measurable by the definition of the measurable-space structure on $G(\Omega \times \Lambda)$, so this follows from the measurability of the graph of multiplication in the space $\mathbb{R}^3$ (equipped with the Borel σ -algebra).

Since S is measurable, we can view it as a measurable subspace of $G(\Omega \times \Lambda)$, and this will also be a standard Borel space. Further, the restriction of f to S is measurable, and since all these spaces are standard Borel, the inverse

$$(2.15) \quad (f|_S)^{-1} : G(\Omega) \times G(\Lambda) \rightarrow S$$

is also measurable. Composing this with the inclusion of S into $G(\Omega \times \Lambda)$, we get a measurable map $G(\Omega) \times G(\Lambda) \rightarrow G(\Omega \times \Lambda)$ that takes a pair of probability measures to their product, as desired. $\square$

We will also use the following proposition, which states that under a standard Borel hypothesis, certain functions that we will care about can be chosen to be measurable.

Proposition 2.10. *Let X and Y be random variables on a standard Borel probability space $(\Omega, \mathbf{P})$, and suppose that there is a function $f: \mathcal{R}X \rightarrow \mathcal{R}Y$ such that $Y = f(X)$ almost everywhere. Then, in fact there is a measurable function $g: \mathcal{R}X \rightarrow \mathcal{R}Y$ satisfying $Y = g(X)$ almost everywhere.*

Proof. By the disintegration theorem 2.4, we can pick a regular conditional probability distribution $x \mapsto \mathbf{P}(\cdot | X = x)$, which is a measurable function $\mathcal{R}X \rightarrow G(\Omega)$. We will construct the desired function $f: \mathcal{R}X \rightarrow \mathcal{R}Y$ by composing this map with a measurable map $T: G(\Omega) \rightarrow \mathcal{R}Y$.

First, let $A \subseteq \Omega$ be the event $Y = f(X)$. Since A has measure 1,

$$(2.16) \quad \int \mathbf{P}(A | X = x) dX_*\mathbf{P}(x) = 1,$$

so, for almost all $x \in \mathcal{R}X$ under the measure $X_*\mathbf{P}$, we have $\mathbf{P}(A | X = x) = 1$. $\square$

Proposition 2.11. *and let $\mathbf{P}(\cdot | X = x)$ be a regular conditional probability distribution on Ω given X . Further, suppose that for almost all $x \in \mathcal{R}X$, there exists $y \in \mathcal{R}Y$ such that $\mathbf{P}(Y = y | X = x) = 1$. In other words, the probability measure $\mathbf{P}(\cdot | X = x)$ is concentrated on some set $Y^{-1}(\{y\})$, with $y \in \mathcal{R}Y$. Then, then there is a measurable function $f: \mathcal{R}X \rightarrow \mathcal{R}Y$ such that $Y = f(X)$ almost everywhere.*

Proof. By the disintegration theorem, we have a measurable map $\mathcal{R}X \rightarrow G(\Omega)$ that maps each $x \in \mathcal{R}X$ to $\mathbf{P}(\cdot | X = x)$. We will construct the desired function $f: \mathcal{R}X \rightarrow \mathcal{R}Y$ by composing this map with a measurable map $T: G(\Omega) \rightarrow \mathcal{R}Y$ with the property that for any $\mathbf{Q} \in G(\Omega)$, if there is some $y \in \mathcal{R}Y$ such that $\mathbf{Q}(Y = y) = 1$, then $T(\mathbf{Q}) = y$. First, we confirm that this property suffices. If; we could compose such maps to obtain a map $f: X \rightarrow \mathcal{R}Y$. $\square$

Corollary 2.12. *Let X and Y be random variables with standard Borel ranges on a standard Borel probability space $(\Omega, \mathbf{P})$ such that $H(Y | X) = 0$. Then, then there is a measurable function $f: \mathcal{R}X \rightarrow \mathcal{R}Y$ such that $Y = f(X)$ almost everywhere.*

Proof. *****We use a regular conditional probability distribution $\mathbf{P}$, as in Definition 2.5. We know ***** that $\mathbf{P}(\cdot | X = x)$ is a Dirac measure in $G(\Omega)$ for almost every $x \in \mathcal{R}X$. *****I'll finish this later. The Dirac measures form a Borel set, which I should maybe derive from the theorem that measurable functions have measurable images. Then, we can modify the function everywhere else to make it into a function $\mathcal{R}X \rightarrow \mathcal{R}Y$. $\square$

2.1. Entropy and mutual information. We want to speak about quantities such as mutual information and conditional mutual information of random variables. Definitions of such quantities in great generality have been given by Wyner (1978), who also developed the theory appropriate to these definitions. In practice, it is fine to assume that all spaces are standard Borel, in which case the theory is somewhat easier, but this work does make it meaningful for us to speak of mutual information without such an assumption. Wyner's definitions are as follows.

Definition 2.13. Let X, Y, Z be random variables on a probability space Ω . Then,

$$(2.17) \quad I(X; Y) = \sup_{\tilde{X}, \tilde{Y}} I(\tilde{X}; \tilde{Y}),$$

where the supremum is taken over all random variables $\tilde{X}, \tilde{Y}$ on Ω such that each of $\tilde{X}$ and $\tilde{Y}$ has finite range, and such that the sub- σ -algebra of Ω induced by $\tilde{X}$ is a subalgebra of that induced by X , and similarly for $\tilde{Y}$ and Y . Note that this expression takes values in the extended positive reals, $[0, \infty]$, where the supremum is meant in the sense of the order on that set. Further,

$$(2.18) \quad I(X; Y | Z) = \sup_{\tilde{X}, \tilde{Y}} I(\tilde{X}, \tilde{Y} | Z).$$

In particular, $I(\tilde{X}, \tilde{Y} | Z)$ is well-defined (though possibly infinite), because of the finite-range condition.

3. COLLECTED DEFINITIONS

I'm not really sure how to handle this. Basically, these definitions are in Eisenstat (2025), but in some cases they're modified by the new definitions of random variable models and latent variable models. Maybe it's sufficient to cite them, but there's some benefit to having them in one place. Maybe this can become a separate supplementary document.

Definition 3.1. A *morphism* of random variable models has the form

$$(3.1) \quad (\pi, \iota, (f_j)_{j \in J}) : (\Omega, (X_i)_{i \in I}) \rightarrow (\Theta, (Y_j)_{j \in J}),$$

where $\pi : \Omega \rightarrow \Theta$ is a probability-preserving map, ι is a function $J \rightarrow I$, and f_j is a function from the range of $X_{\iota(j)}$ to the range of Y_j . We require that for all $j \in J$, we have $Y_j = f_j(X_{\iota(j)})$ almost everywhere on Ω . Note that the function f_j is automatically measurable, since these random variables have countable and discrete range, and from the same premises that the condition $Y_j = f_j(X_{\iota(j)})$ defines a measurable set.

Definition 3.2. Given two morphisms

$$(3.2) \quad (\pi, \iota, (f_j)_{j \in J}) : (\Omega, (X_i)_{i \in I}) \rightarrow (\Theta, (Y_j)_{j \in J})$$

$$(3.3) \quad (\rho, \nu, (g_k)_{k \in K}) : (\Theta, (Y_j)_{j \in J}) \rightarrow (\Pi, (Z_k)_{k \in K}),$$

their *composite* is the morphism

$$(3.4) \quad (\rho \circ \pi, \iota \circ \nu, (g_k \circ f_{\nu(k)})_{k \in K}) : (\Omega, (X_i)_{i \in I}) \rightarrow (\Pi, (Z_k)_{k \in K}).$$

4. RANDOM VARIABLE MODELS AND LATENT VARIABLE MODELS

In this section we'll discuss the basic theory of random variable models, latent variable models, objectivity, and almost perfect condensation. These ideas are present in Eisenstat (2025), but the treatment here will generalize them somewhat and adapt them to our further discussion.

A *random variable model* will represent a joint distribution over a set of variables, and a *latent variable model* for a random variable model will introduce a set of latent variables intended to organize the structure of the given random variable model. The definitions here are more general than those in Eisenstat (2025), since they are more permissive of continuous probability measures, where entropies can be infinite.

Definition 4.1. A *random variable model* is a pair $(\Omega, (X_i)_{i \in I})$, where Ω is a standard Borel probability space, and $(X_i)_{i \in I}$ is a finite family of random variables $X_i: \Omega \rightarrow \mathcal{R} X_i$, each with standard Borel range.

Definition 4.2. A *latent variable model* associated with a random variable model $(\Omega, (X_i)_{i \in I})$ consists of a random variable model $(\Lambda, (Y_A)_{A \in \mathcal{J}})$ (I have to decide whether to use a *contribution relation* or a *power set*), together with a probability-preserving map $\pi: \Lambda \rightarrow \Omega$ such that for each $i \in I$, the random variable X_i is almost everywhere a function of those random variables Y_A such that $i \in A$.

In order to state the objectivity theorem, we need the definitions of *morphism*, *amalgamation*, *polar*, and *intersection tree* from Eisenstat (2025). Some of the accompanying theorems, and maybe even the definitions, have to be adapted to this generality.

5. OBJECTIVITY INEQUALITIES

We can strengthen the objectivity theorem, Theorem 6.8 of Eisenstat (2025) in two ways. First, we'll use this new, more general sense of latent variable models, where variables may take values in standard Borel measurable spaces and may have infinite entropy. Second, we can give a tighter bound using the idea of *independent amalgamation*, which we'll discuss. While later sections have enough conceptual independence of these ideas that they could also be developed along similar lines with just what's in Eisenstat (2025), we'll choose to instead use these more general ideas.

Definition 5.1. Let $\mathcal{X} = (\Omega, (X_i)_{i \in I})$ be a random variable model, and let $\mathcal{Y} = ((\Lambda_1, (Y_A)_{A \in \mathcal{J}}), \pi_1)$ and $\mathcal{Z} = ((\Lambda_2, (Z_B)_{B \in \mathcal{K}}), \pi_2)$ be two associated latent variable models. An *independent amalgamation*

It will work better to write the main theorems first and let that guide us as to the simplest definitions

In the sequel, inequalities should be read in the sense of the extended reals. That is, if we write $\text{LHS} \leq \text{RHS}$, then we do not assert that LHS and RHS denote real numbers, but instead that they denote extended real numbers—elements of $\{-\infty\} \cup \mathbb{R} \cup \{\infty\}$ —that stand in the natural order relation to each other.

Lemma 5.2 is a generalization of Lemma 6.4 from Eisenstat (2025), which uses mutual information instead of entropy. The ideas are roughly the same, but this generalization is useful for cases where the distribution of some relevant variable

is continuous, and thus the entropy is infinite, or otherwise if the entropy is very large but better bounds are still possible using mutual information.

Lemma 5.2. *Let X, Y, Z_1, Z_2 , and C be random variables on a probability space, and suppose that (Z_1, Z_2, C) is independent of Y given X . Then,*

$$(5.1) \quad I(Y; X | C) \leq I(Y; X | Z_1, C) + I(Y; X | Z_2, C) + I(Z_1; Z_2 | C).$$

Further, we have the exact statement

$$(5.2) \quad \begin{aligned} I(Y; X | C) + I(Y; X | Z_1, Z_2, C) + I(Z_1; Z_2 | Y, C) \\ = I(Y; X | Z_1, C) + I(Y; X | Z_2, C) + I(Z_1; Z_2 | C). \end{aligned}$$

Proof. We will have to calculate, following a similar idea as in Lemma 6.4 of Eisenstat (2025). First, we can expand the terms on both sides of (5.2), using the conditional independence hypothesis. We have

$$(5.3) \quad I(Y; X | C) = I(Y; X, Z_1, Z_2 | C)$$

and, for (i, j) either $(1, 2)$ or $(2, 1)$,

$$(5.4) \quad I(Y; X | Z_i, C) = I(Y; X, Z_j | Z_i, C),$$

so

$$\begin{aligned} I(Y; X | C) &= I(Y; Z_1, Z_2 | C) + I(Y; X | Z_1, Z_2, C) \\ I(Y; X | Z_i, C) &= I(Y; Z_j | Z_i, C) + I(Y; X | Z_1, Z_2, C). \end{aligned}$$

Now, since all the following terms are nonnegative,

$$\begin{aligned} I(Y; Z_2 | Z_1, C) + I(Y; Z_1 | Z_2, C) + I(Z_1; Z_2 | C) \\ = I(Y; Z_2 | Z_1, C) + I(Y, Z_2; Z_1 | C) \\ = I(Y; Z_2 | Z_1, C) + I(Y; Z_1 | C) + I(Z_2; Z_1 | Y, C) \\ = I(Y; Z_1, Z_2 | C) + I(Z_1; Z_2 | Y, C), \end{aligned}$$

so

$$\begin{aligned} I(Y; X | Z_1, C) + I(Y; X | Z_2, C) + I(Z_1; Z_2 | C) \\ = 2I(Y; X | Z_1, Z_2, C) + I(Y; Z_1, Z_2 | C) + I(Z_1; Z_2 | Y, C) \\ = I(Y; X | C) + I(Y; X | Z_1, Z_2, C) + I(Z_1; Z_2 | Y, C). \end{aligned}$$

The desired inequality follows immediately. $\square$

The real content of this theorem is not using that $Y_{\supseteq A}$ is $Y_{\supseteq A}$; it is true for any variable. The actual $Y_{\supseteq A}$ part should maybe only come in APC?

Theorem 5.3. *Let $\mathcal{X}$ be a random variable model with random variables $(X_i)_{i \in I}$, and let $\mathcal{Y}$ and $\mathcal{Z}$ be two associated latent variable models with latent variables $(Y_A)_{A \in \mathcal{J}}$ and $(Z_B)_{B \in \mathcal{K}}$, respectively. Consider any independent amalgamation of $\mathcal{Y}$ and $\mathcal{Z}$; in the sequel, we will abuse notation by writing variables X, Y , and Z when we mean their pullbacks to the amalgamated probability space. Now, let $\mathcal{F} \subseteq \mathcal{K}$, and pick any intersection tree T , whose leaves have labels comprising the set $\mathcal{G}$, where $\mathcal{G} \subseteq \mathcal{P}I$, and whose internal nodes form a set N , such that $\mathcal{F}$ contains the intersection of the polar $\mathcal{G}^\circ$ with $\mathcal{K}$. Then,*

$$(5.5) \quad I(Y_{\supseteq A}; X | Z_{\mathcal{F}}) \leq \sum_{B \in \mathcal{G}} I(Y_{\supseteq A}; X | X_B) + \sum_{v \in N} I(Z_{\mathcal{L}(v)}; Z_{\mathcal{R}(v)} | Z_{\mathcal{I}(v)}).$$

Proof. We use the same inductive idea as in Theorem 6.8 of Eisenstat (2025); for each $v \in T$, we want to show that

$$(5.6) \quad I(Y_{\supseteq A}; X \mid Z_{\mathcal{I}(v)}) \leq \sum_{B \in L(v)} I(Y_{\supseteq A}; X \mid X_B) + \sum_{v \in N} I(Z_{\mathcal{L}(v)}; Z_{\mathcal{R}(v)} \mid Z_{\mathcal{I}(v)}).$$

If v is a leaf, then $\mathcal{I}(v) = \{C : C \cap \ell(v) \neq \emptyset\}$, so $X_{\ell(v)}$ is determined by $Z_{\mathcal{I}(v)}$, and so,

$$(5.7) \quad I(Y_{\supseteq A}; X \mid Z_{\mathcal{I}(v)}) \leq I(Y_{\supseteq A}; X \mid X_{\ell(v)}),$$

which is the base case. If v is internal, suppose that both its children satisfy this inductive hypothesis. Using Lemma 5.2 and the assumption of independent amalgamation, and writing $L(v)$ for the leaves and $N(v)$ for the internal nodes of the subtree of T whose root is v ,

$$\begin{aligned} I(Y_{\supseteq A}; X \mid Z_{\mathcal{I}(v)}) &\leq I(Y_{\supseteq A}; X \mid Z_{\mathcal{L}(v)}) + I(Y_{\supseteq A}; X \mid Z_{\mathcal{R}(v)}) + I(Z_{\mathcal{L}(v)}; Z_{\mathcal{R}(v)} \mid Z_{\mathcal{I}(v)}) \\ &= \sum_{B \in L(v)} I(Y_{\supseteq A}; X \mid X_B) + \sum_{v \in N(v)} I(Z_{\mathcal{L}(v)}; Z_{\mathcal{R}(v)} \mid Z_{\mathcal{I}(v)}). \end{aligned}$$

In particular, at the root v_0 of T , since $\mathcal{F} \supseteq \mathcal{I}(v_0)$,

$$\begin{aligned} I(Y_{\supseteq A}; X \mid Z_{\mathcal{F}}) &\leq I(Y_{\supseteq A}; X \mid Z_{\mathcal{I}(v_0)}) \\ &\leq \sum_{B \in \mathcal{G}} I(Y_{\supseteq A}; X \mid X_B) + \sum_{v \in N} I(Z_{\mathcal{L}(v)}; Z_{\mathcal{R}(v)} \mid Z_{\mathcal{I}(v)}) \end{aligned}$$

□

6. ALMOST PERFECT CONDENSATION

In order to apply Theorem 5.3, we want to place appropriate hypotheses on latent variable models. *****

Definition 6.1. Let $(X_i)_{i \in I}$ be the random variables of a random variable model, and let $(Y_A)_{A \in \mathcal{J}}$ be the latent variables of an associated latent variable model $\mathcal{Y}$. We say that $\mathcal{Y}$ has (r, ε) -reconstruction if for all sets $A \in \mathcal{J}$ and $B \in \mathcal{PI}$ such that $|A \cap B| > r$, we have the inequality $H(Y_{\supseteq A} \mid X_B) \leq \varepsilon$. Also, $\mathcal{Y}$ has the (m, ε) -ordered Markov condition if

- (1) $|\mathcal{J}| \leq m$ and
- (2) for all upward-closed sets $\mathcal{F}, \mathcal{G} \subseteq \mathcal{J}$, we have

$$(6.1) \quad I(Y_{\mathcal{F}}; Y_{\mathcal{G}} \mid Y_{\mathcal{F} \cap \mathcal{G}}) \leq \varepsilon.$$

Theorem 6.2. Let $(X_i)_{i \in I}$ be the random variables of a random variable model, and let $\mathcal{Y}$ and $\mathcal{Z}$ be two associated latent variable models with latent variables $(Y_A)_{A \in \mathcal{J}}$ and $(Z_B)_{B \in \mathcal{K}}$. Suppose that $\mathcal{Y}$ has $(\preceq, \varepsilon)$ -reconstruction, and that $\mathcal{Z}$ is (m, η) -Markov. Then, for any independent amalgamation of $\mathcal{Y}$ and $\mathcal{Z}$, we have

$$(6.2) \quad \text{*****}.$$

$$(6.3) \quad I(Y_{\mathcal{F}} \mid).$$

Proof. Let $\mathcal{G} = B_r(\mathcal{F})$, and define $\mathcal{H} = \mathcal{K} - \mathcal{G}$. □

Corollary 6.3. *Let $(X_i)_{i \in I}$ be the random variables of a random variable model, and suppose we have two associated latent variable models with latent variables $(Y_A)_{A \in \mathcal{J}}$ and $(Z_B)_{B \in \mathcal{K}}$, satisfying $(\preceq, \varepsilon)$ -reconstruction, and the (m, η) -Markov condition, respectively. Then, for any independent amalgamation of $\mathcal{Y}$ and $\mathcal{Z}$, we have*

$$(6.4) \quad I(Y_{\mathcal{F}}; X \mid Z_{B_r(\mathcal{F})}) \leq m\varepsilon + m\eta.$$

Proof. Let □

7. EXAMPLES

8. ALGORITHMIC INFORMATION CONDENSATION

The theory of condensation admits a translation, relatively straightforward in its basic aspects, from a setting of probability spaces, random variables, and Shannon entropy, to a setting of strings and string complexity measures, such as the various kinds of Kolmogorov complexity. In this section, we will develop these ideas, in a way paralleling the development for random variable models and latent variable models.

The main motivation behind this translation is to better meet the actual setting of learning. A random variable model is a joint distribution on a set of variables. While we often make assumptions of independent and identically distributed data in learning theory, this is far from describing many actual situations of learning. Even when such assumptions are warranted, such a distribution must be estimated from data; we ought not separate the stages of estimating a distribution from data and then understanding the distribution, when we could instead try to understand the data.

If this joint distribution is interpreted as expressing some sort of propensities of some random events¹, then agents are often in situations in which there may or may not be any meaningful propensities, and there are at least no reasonable knowable propensities. A major class of exceptions is when the propensities are related to frequencies of repeating events.

We will treat the basic objects as strings and focus our analysis to complexity measures appropriate to strings, rather than working in a fully abstract setting of “information-bearing objects” and abstract “information measures”. Why is this simpler? We don’t really have other interesting examples, though you could take e.g. fields as information objects and the degree of field extensions as conditional complexity.

REFERENCES

- Eisenstat, Sam (2025). “Condensation: a theory of concepts.” In: *ODYSSEY 2025 Conference*.
- Fritz, Tobias (Aug. 26, 2020). “A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics.” In: *Advances in Mathematics* 370, pp. 107–239.
- Giry, Michèle (1982). “A categorical approach to probability theory.” In: *Categorical Aspects of Topology and Analysis*. Ed. by B. Banaschewski. Berlin, Heidelberg: Springer, pp. 68–85.

¹I mean to encompass a variety of metaphysical accounts of what a propensity is here.

- Kechris, Alexander S. (1995). *Classical Descriptive Set Theory*. Vol. 156. Graduate Texts in Mathematics. New York, NY: Springer.
- Mackey, George W. (1957). “Borel structure in groups and their duals.” In: *Transactions of the American Mathematical Society* 85.1, pp. 134–165.
- Wyner, A. D. (July 1, 1978). “A definition of conditional mutual information for arbitrary ensembles.” In: *Information and Control* 38.1, pp. 51–59. ISSN: 0019-9958. DOI: 10.1016/S0019-9958(78)90026-8. URL: <https://www.sciencedirect.com/science/article/pii/S0019995878900268> (visited on 07/09/2026).